AI & MACHINE LEARNING

Internship Program

Project Brief & Guidelines

Python	Pandas	Scikit-learn	TensorFlow

Fully Remote · Self-Paced · Certificate on Completion

Two Real AI/ML Projects · Python-Based · Verified Certificate Included

plutoacademy.in · MSME: UDYAM-RJ-06-0057331

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Welcome to Pluto Academy

Congratulations on joining the **Pluto Academy AI & Machine Learning Internship Program**. This is a fully remote, 1-month hands-on program where you will work on two complete AI/ML projects using Python — covering data analysis, machine learning model building, evaluation, and deployment. You will work with real datasets and industry-standard tools.

This is not a tutorial. You receive a problem statement and build the solution from scratch — exactly the way it works at a real AI/ML team. By the end, you will have two Python projects on GitHub that any employer can review.

Program Detail	Information
Program Name	AI & Machine Learning Internship
Tech Stack	Python · Pandas · NumPy · Matplotlib · Scikit-learn · TensorFlow
Duration	1 Month — Fully Remote
Projects	2 (both mandatory — submit together at the end)
Fee	Rs. 99 (one-time, no hidden charges)
Certificate	Issued within 48 hours of successful evaluation
Submission Form	forms.gle/AVjDk51he5zYAHWY7
Verify Certificate	plutoacademy.in/verify-certificates/
Support	WhatsApp +91 7597129727 · plutoacademy1527@gmail.com
Join Here	aimlInternship.plutoacademy.in

IMPORTANT: Both projects are mandatory. Submit both together via the Google Form at the end of your 1-month internship. Do not submit them separately.

PRE-REQUISITE: Basic knowledge of Python is recommended — variables, loops, functions. If you have completed our AI/ML Course, you are fully prepared for this internship.

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Tools & Setup (All Free)

Every tool used in this internship is completely free. You do not need to install anything locally — Google Colab runs Python in your browser with GPU support.

Google Colab — https://colab.research.google.com
Free browser-based Python environment with GPU — no installation needed. Recommended for all projects.

Python — https://python.org
Core programming language for all AI/ML work.

Pandas & NumPy	https://pandas.pydata.org Data manipulation and numerical computing libraries — pre-installed on Colab.
Matplotlib/Seaborn	https://matplotlib.org Data visualization libraries for charts, plots, and EDA visualizations.
Scikit-learn	https://scikit-learn.org ML library for training, evaluating, and comparing models — pre-installed on Colab.
TensorFlow/Keras	https://tensorflow.org Deep learning framework — used for neural network project. Pre-installed on Colab.
Kaggle	https://kaggle.com Download real-world datasets for both projects. Create a free account.
GitHub	https://github.com Host your code. Both project repos must be public before submitting.

PROJECT 01

Exploratory Data Analysis & Insights Report

EDA	DATA VISUALIZATION	COMPLETE BY WEEK 2

The Problem Statement

You are a Data Analyst hired at a company. Your manager has given you a real-world dataset and asked you to understand the data, find patterns and trends, visualize the key insights, and write a clear report with 5 business or research recommendations based on your findings.

Choose ONE dataset from Kaggle:

- ◆ Netflix Movies & TV Shows Dataset — https://www.kaggle.com/datasets/shivamb/netflix-shows
- ◆ World Happiness Report Dataset — https://www.kaggle.com/datasets/unsdsn/world-happiness
- ◆ Global COVID-19 Dataset — https://www.kaggle.com/datasets/josephassaker/world-covid19-dataset
- ◆ Zomato Restaurant Dataset — https://www.kaggle.com/datasets/shrutimehta/zomato-restaurants-data

What You Must Build

1 **Step 1 — Load & Inspect the Data**
Use Pandas to load the CSV, check shape, data types, missing values, and duplicates. Write a 5-line summary of what the data contains.

2 **Step 2 — Clean the Data**
Handle missing values (drop or fill), fix data types, remove irrelevant columns. Document every cleaning decision and why you made it.

3 **Step 3 — Exploratory Data Analysis**
Answer 5 specific questions about the data using groupby, value_counts, describe, and other Pandas methods. Each question must have a code output.

4 **Step 4 — Visualizations (Minimum 6 Charts)**
Create at least 6 different chart types: bar chart, line chart, histogram, scatter plot, pie chart, heatmap. All charts must have titles, axis labels, and be readable.

5 **Step 5 — Insights Report**
Write 5 numbered business or research insights based purely on your analysis. Each insight must reference a specific chart or finding.

Tools for Project 01

Python	Pandas	NumPy	Matplotlib	Seaborn	Colab

What to Submit for Project 01

- ✓ GitHub repository link (public repo, notebook file .ipynb, and a README)
- ✓ Google Colab notebook link (make sure sharing is set to 'Anyone with link can view')

- ✓ All 6+ visualizations must be visible in the notebook output
- ✓ Insights Report — 5 numbered insights written in plain English inside the notebook
- ✓ Short note (3–5 lines) on which finding surprised you the most

Evaluation — 100 Marks

Parameter	Marks	What We Look For
Data Cleaning Quality	20	Proper handling of missing values, type fixes, documentation
EDA Depth (5 questions)	25	Relevant, well-coded analysis using Pandas methods
Visualization Quality	25	6+ correct chart types, titles, labels, readability
Insights Report	20	5 clear, data-backed insights in plain English
GitHub & Notebook	10	Public repo, clean code, readable notebook
TOTAL — pass = 60+	**100**	**Minimum 60 required to receive certificate**

PROJECT 02

Machine Learning Model — Predict & Evaluate

ML MODEL | SCIKIT-LEARN | COMPLETE BY WEEK 4

The Problem Statement

You are an ML Engineer at a tech company. Your task is to build, train, and evaluate a machine learning model that predicts an outcome from real-world data. You must compare at least 3 different ML algorithms, evaluate them using proper metrics, and clearly explain which model performed best and why.

Choose ONE dataset from Kaggle:

- ◆ Titanic — Survival Prediction (Classification) — https://www.kaggle.com/c/titanic
- ◆ House Prices — Sale Price Prediction (Regression) — https://www.kaggle.com/c/house-prices-advanced-regression-techniques
- ◆ Heart Disease Prediction (Classification) — https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset
- ◆ Iris Flower Classification (Beginners welcome) — https://www.kaggle.com/datasets/uciml/iris

What You Must Build

1 **Step 1 — Load, Explore & Preprocess**

Load the dataset, check for missing values, encode categorical variables, and split into train/test sets (80/20). Explain every preprocessing decision.

2 **Step 2 — Feature Engineering**

Identify which features matter most using correlation analysis or feature importance. Drop irrelevant features and explain why.

3 **Step 3 — Train 3 Different Models**

Train at least 3 algorithms: e.g. Logistic Regression + Random Forest + KNN (classification) or Linear Regression + Decision Tree + Random Forest (regression).

4 **Step 4 — Evaluate & Compare All Models**

Evaluate all 3 models using proper metrics (Accuracy, Precision, Recall, F1 for classification; RMSE, MAE, R^2 for regression). Show results in a clear comparison table.

5 **Step 5 — Best Model Analysis & Conclusion**

Identify the best performing model. Explain why it performed better. Show a confusion matrix (classification) or residual plot (regression). Write a 5-line conclusion.

What to Submit for Project 02

- ✓ GitHub repository link (public, with README and dataset source mentioned)
- ✓ Google Colab notebook link (sharing: 'Anyone with link can view')
- ✓ Comparison table of all 3 models with evaluation metrics clearly shown

✓ Confusion matrix or residual plot for the best model
✓ 5-line written conclusion explaining your best model choice

Evaluation — 100 Marks

Parameter	Marks	What We Look For
Preprocessing & Feature Eng.	20	Clean pipeline, proper encoding, logical feature selection
Model Training (3 models)	25	All 3 trained correctly with proper train/test split
Evaluation & Comparison Table	25	Correct metrics, comparison table clear and complete
Best Model Analysis	20	Confusion matrix/residual plot, strong conclusion
GitHub & Code Quality	10	Clean code, public repo, meaningful comments
TOTAL — pass = 60+	**100**	**Minimum 60 required to receive certificate**

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Suggested 4-Week Timeline

You have one full month. Here is the recommended week-by-week plan to complete both projects comfortably.

WEEK 1 **Setup + Project 01 Start**

• Create Google Colab account, connect to Google Drive • Download your chosen dataset from Kaggle • Load, inspect, and clean the dataset using Pandas • Answer the first 2–3 EDA questions with code • Join the Telegram support channel for guidance

WEEK 2 **Complete & Finalise Project 01**

• Answer remaining EDA questions • Create all 6+ visualizations with proper titles and labels • Write the 5-point Insights Report inside the notebook • Push clean code to a public GitHub repo with README • Share Colab notebook link with 'Anyone can view' permission

WEEK 3 **Project 02 — Build Phase**

• Choose your ML dataset and load it in a new Colab notebook • Preprocess: handle missing values, encode categoricals, split data • Perform feature engineering and correlation analysis • Train Model 1 and Model 2 — record evaluation metrics

WEEK 4 **Complete, Evaluate & Submit**

• Train Model 3 and create the full comparison table • Generate confusion matrix or residual plot for best model • Write the 5-line conclusion • Push everything to GitHub with a clean README • Submit both project links via Google Form • WhatsApp +91 7597129727 after submitting — certificate in 48 hours

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How to Submit

ONE SUBMISSION FOR BOTH PROJECTS: Submit both at the end of your 1-month internship via a single Google Form. Include all GitHub and Colab links together.

What to include in your submission:

✓ Project 01 — GitHub repo link (public)
✓ Project 01 — Google Colab notebook link (Anyone can view)
✓ Project 02 — GitHub repo link (public)
✓ Project 02 — Google Colab notebook link (Anyone can view)
✓ Project 02 — Screenshot of your 3-model comparison table
✓ Your full name and enrollment email address
✓ Brief note (3–5 lines) on the most interesting finding from each project

1	**Organise your work** Both GitHub repos public, both Colab notebooks shareable. Test all links before submitting.
2	**Fill the Google Form** Open the submission form link (shared on Telegram and in your welcome email). Fill all fields.
3	**Submit the form** You will receive an automatic confirmation email. Save it as proof.
4	**WhatsApp after submitting** Send a WhatsApp to +91 7597129727 with your name. This starts the evaluation process.
5	**Receive your certificate** Evaluated within 48 hours. Certificate sent to your email and WhatsApp.

RESUBMISSION POLICY: Score below 60/100 → you receive detailed feedback and ONE chance to resubmit within 5 days. Score 60+ on resubmission → certificate issued. No further resubmissions.

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DOs and DON'Ts

DO	DON'T
• Write and run your own code in Colab	• Copy-paste notebooks from Kaggle or GitHub
• Use Google and Stack Overflow to learn	• Use AI to write your entire analysis
• Ask questions in the Telegram support channel	• Submit private/inaccessible GitHub repos
• Commit code regularly to GitHub	• Submit without running the notebook first
• Test all links before submitting	• Submit only one project
• Write clear markdown comments in your notebook	• Leave notebooks with errors unresolved
• Submit both project links together	• Miss the 1-month deadline without informing us

ACADEMIC INTEGRITY: You may use AI tools to understand concepts and debug errors. But your analysis, code, and insights must be your own work. Submitting copied notebooks will be immediately disqualified.

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Support & Contact

Stuck on preprocessing? Can't figure out model evaluation? Reach out — we are here to help.

Channel	Details
Telegram	t.me/plutoverse7 — all announcements, updates, doubt support
Email	plutoacademy1527@gmail.com
WhatsApp	+91 7597129727 — message after submitting projects

Instagram	@pluto.academyy · @pluto.careers
YouTube	youtube.com/@PlutoAcademyy — free AI/ML tutorials
Website	plutoacademy.in

We want every student to succeed in this internship. Building real AI/ML projects is challenging — but that challenge is exactly what makes them valuable on your resume. Push through, build something real, and come out with two AI/ML projects that no one can take away from you.

Good luck — we're rooting for you!

— Tishant Agrawal, Founder, Pluto Academy